

Chapter 16 Homework

Part A multiple choice (select the one that is best in each case. 1 point/question)

- What is the pH of a 0.40 M solution of benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, 25 °C? (the $K_a = 6.3 \times 10^{-5}$ for benzoic)
A) 2.30
B) 2.10
C) 1.90
D) 4.20
E) 4.60
- In which of the following aqueous solutions does the weak acid exhibit the lowest percentage ionization?
A) 0.01 M H_2SO_3 ($K_a = 1.4 \times 10^{-2}$)
B) 0.01 M HCN ($K_a = 6.2 \times 10^{-10}$)
C) 0.01 M H_2CO_3 ($K_a = 4.5 \times 10^{-7}$)
D) 0.01 M $\text{HC}_3\text{H}_5\text{O}_2$ ($K_a = 1.3 \times 10^{-5}$)
E) 0.01 M HOCl ($K_a = 3.5 \times 10^{-8}$)
- Which of the following acids, if in solutions of equal concentration, is the most acidic? Assume all act as monoprotic acids.
A) boric acid, $K_a = 5.8 \times 10^{-10}$
B) phenol, $K_a = 1.3 \times 10^{-10}$
C) lactic acid, $K_a = 1.4 \times 10^{-4}$
D) saccharin, $K_a = 2.1 \times 10^{-12}$
E) All of these acids are equally acidic because they are all of equal concentration
- Which of the following is/are strong acid(s)?
(i) HNO_3 (ii) HF (iii) HCl (iv) H_2S
A) (i) and (iv) only B) (i) only C) (iii) only D) (i) and (iii) only E) (ii) and (iii) only
- Given that HClO_4 is a strong acid, how would you expect the acidity or basicity of ClO_4^- ?
A) strong acid B) weak acid C) strong base D) weak base E) negligible basicity
- Which one of the following *cannot* act as a Lewis base?
A) Cr^{3+} B) F^- C) NH_3 D) Cl^- E) CN^-
- In a certain acidic solution at 25 °C, $[\text{H}^+]$ is 10^6 times greater than $[\text{OH}^-]$. What is the value for $[\text{OH}^-]$ for the solution?
A) $1.0 \times 10^{-7} \text{ M}$

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- B) $1.0 \times 10^{-10} M$
C) $1.0 \times 10^{-6} M$
D) $1.0 \times 10^{-2} M$
E) $1.0 \times 10^{-9} M$
8. A 0.1 M aqueous solution of _____ will produce an basic solution at 25.0 °C.
LiF RbBr NaClO₄ NH₄Cl
A) RbBr and NaClO₄
B) LiF and RbBr
C) LiF only
D) NaClO₄ only
E) NH₄Cl only
9. The acid-dissociation constants of sulfurous acid (H₂SO₃) are $K_{a1} = 1.7 \times 10^{-2}$ and $K_{a2} = 6.4 \times 10^{-8}$ at 25.0 °C. Calculate the pH of a 0.163 M aqueous solution of sulfurous acid.
A) 4.53 B) 6.21 C) 1.86 D) 1.35 E) 1.93
10. The hydride ion, H⁻, is a stronger base than the hydroxide ion, OH⁻. The product(s) of the reaction of hydride ion with water is/are _____.
A) H₃O⁺(aq)
B) H₂O₂(aq)
C) OH⁻(aq) + 2 H⁺(aq)
D) no reaction occurs
E) OH⁻(aq) + H₂(g)
11. The conjugate acid of CH₃NH₂ is _____.
A) CH₃NH₃⁺ B) CH₃NH₂ C) CH₃NH₂⁺ D) CH₃NH⁺ E) none of the above
12. In the gas phase reaction below, NH₃ is acting as a(n) _____ base but not as a(n) _____ base.
- $$\begin{array}{c} \text{H} - \ddot{\text{N}} - \text{H} \\ | \\ \text{H} \end{array} + \text{H}^+ \longrightarrow \begin{array}{c} \text{H} \\ | \\ \text{H} - \text{N}^+ - \text{H} \\ | \\ \text{H} \end{array}$$
- A) Arrhenius, Brønsted-Lowry
B) Lewis, Arrhenius
C) Brønsted-Lowry, Lewis
D) Lewis, Brønsted-Lowry

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E) Arrhenius, Lewis

13. An aqueous solution contains 0.050 M of methylamine (CH_3NH_2). The concentration of H^+ in this solution is _____ M. K_b for methylamine is 4.4×10^{-4} .
- A) 0.050 B) 2.2×10^{-13} C) 2.2×10^{-12} D) 4.5×10^{-13} E) 2.9×10^{-13}
14. The base-dissociation constant, K_b , for pyridine, $\text{C}_5\text{H}_5\text{N}$, is 1.5×10^{-9} at 298K. The acid-dissociation constant, K_a , for the pyridinium ion, $\text{C}_5\text{H}_5\text{NH}^+$, is _____.
- A) 1.0×10^{-7} B) 1.4×10^{-23} C) 7.1×10^{-4} D) 6.7×10^{-6} E) 1.4×10^{-5}
15. Arrange the following substances in order from weakest to strongest acid: KH, HBr, AsH_3 , H_2Se , HI
- A) $\text{AsH}_3 < \text{H}_2\text{Se} < \text{KH} < \text{HBr} < \text{HI}$
B) $\text{H}_2\text{Se} < \text{HI} < \text{HBr} < \text{AsH}_3 < \text{KH}$
C) $\text{H}_2\text{Se} < \text{AsH}_3 < \text{HBr} < \text{HI} < \text{KH}$
D) $\text{KH} < \text{HI} < \text{HBr} < \text{AsH}_3 < \text{H}_2\text{Se}$
E) $\text{KH} < \text{AsH}_3 < \text{H}_2\text{Se} < \text{HBr} < \text{HI}$

Part B Short question (5 points)

16. Benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) and aniline ($\text{C}_6\text{H}_5\text{NH}_2$) are both derivatives of benzene. Benzoic acid is an acid with $K_a = 6.3 \times 10^{-5}$ and aniline is a base with $K_b = 4.3 \times 10^{-10}$ at 25 °C.
- (a) What are the conjugate base of benzoic acid and the conjugate acid of aniline?
- (b) Anilinium chloride ($\text{C}_6\text{H}_5\text{NH}_3\text{Cl}$) is a strong electrolyte that dissociates into anilinium ions ($\text{C}_6\text{H}_5\text{NH}_3^+$) and chloride ions. Which will be more acidic, a 0.10 M solution of benzoic acid or a 0.10 M solution of anilinium chloride?
- (c) What is the value of the equilibrium constant for the following equilibrium at 25 °C?
- $$\text{C}_6\text{H}_5\text{COOH}(aq) + \text{C}_6\text{H}_5\text{NH}_2(aq) \rightleftharpoons \text{C}_6\text{H}_5\text{COO}^-(aq) + \text{C}_6\text{H}_5\text{NH}_3^+(aq)$$